MODULE 01 / HANDS-ON PRACTICE

ASSIGNMENT 01
Setup & Variables

TypeScript for Playwright Automation | Beginner Level

LEARNING OBJECTIVES

Configure a working TypeScript project in VS Code, understand var, let, const, practice assignment and reassignment, and format console output confidently.

LEVEL	QUESTIONS	SUBMISSION
Beginner	12 multi-part tasks	.ts files + screenshot

HOW TO COMPLETE

1. Set up the project using the commands on the next page.
2. Create src/assignment1.ts and complete all coding tasks.
3. Before running each question, predict the output and explain why.
4. Add comments separating each question and submit your code plus terminal output.

NO SOLUTIONS PROVIDED • INDEPENDENT STUDENT PRACTICE

PART A / ENVIRONMENT SETUP

Get your tools ready

Use the VS Code integrated terminal. On Windows, Command Prompt is convenient if PowerShell blocks npm.ps1 scripts.

01 Check installations

```
node -v
npm -v
npx -v
```

02 Create and initialize project

```
mkdir typescript-assignment-1
cd typescript-assignment-1
npm init -y
```

03 Install TypeScript and generate config

```
npm install -D typescript
npx tsc --version
npx tsc --init
```

04 Edit tsconfig.json

```
{
  "compilerOptions": {
    "target": "ES2020",
    "module": "CommonJS",
    "strict": true,
    "rootDir": "./src",
    "outDir": "./dist"
  },
  "include": ["src/**/*"]
}
```

05 Create src/assignment1.ts, then compile and run

```
npx tsc
node dist/assignment1.js
```

Checkpoint: Identify the purpose of Node.js, npm, npx, tsc, package.json, node_modules, and tsconfig.json.

PART B / CODING CHALLENGES

12 real-world practice tasks

Complete every subquestion. Keep each task in its own commented section. Do not move to the next task until your code compiles successfully.

Q01 Meet the Student

Create variables for a new student joining the course.

a. Store student name, age, city and course title in separate variables.

b. Use const for values that will not change and let for age if it can be updated.

c. Print all four values using a single console.log().

d. Print a 4-line student profile using \n in one console.log().

e. Change city and explain why reassignment is allowed or not.

Q02 Shopping Checkout

A customer is buying 3 notebooks priced at Rs. 45 each.

a. Declare the notebook price and quantity with appropriate keywords.

b. Calculate the total cost and print it with a readable label.

c. Update quantity to 5 and print the new bill.

d. Add a fixed Rs. 20 delivery charge; calculate the final amount.

e. Explain which variables should be const and which should be let.

Q03 Monthly Salary Revision

An employee receives a salary increase and a fixed bonus.

a. Create a variable with an initial salary of Rs. 30000.

b. Reassign salary after a Rs. 4000 increment.

c. Create a constant bonus of Rs. 2000.

d. Calculate and print salary plus bonus.

e. Print both old and new salary on separate lines with one console.log().

Q04 Exam Scorecard

A student has marks in Mathematics, Science and English.

a. Store marks 78, 89 and 84 as three variables.

b. Calculate the total marks out of 300.

c. Calculate percentage and show it using a template literal.

d. Change the Science marks to 94 and calculate the new percentage.

e. Predict the outputs before running.

Q05 Swap Challenge

Two variables hold different values.

a. Declare a = 10 and b = 20 using let.

b. Print both values before swapping.

c. Swap them using a third variable.

d. Print both values after swapping in a single console.log().

e. Challenge: repeat without using a third variable.

Q06 Login Test Data

Prepare fictional data for an automated login test.

a. Store an application URL and username in constants.

b. Store login attempt count in a let variable starting at 0.

c. Increase the attempt count twice.

d. Print the username, URL and attempts using a multiline template literal.

e. Explain why passwords should not be logged in real automation tests.

Q07 Ticket Booking

A booking counter sells tickets at a fixed price.

a. Set ticket price to Rs. 250 and initial ticket quantity to 2.

b. Calculate and print the initial payable amount.

c. Update ticket quantity to 4, then recalculate.

d. Add a one-time convenience fee of Rs. 30.

e. Print a neatly formatted invoice using only one console.log().

Q08 Predict the Output

Study code without running it first.

a. Predict: `let x = 5; x = x + 10; console.log(x);`

b. Predict: `const rate = 8; console.log(rate * 3);`

c. Predict: `var city = "A"; var city = "B"; console.log(city);`

d. Explain whether `const count = 1; count = 2;` compiles successfully.

e. Run each valid snippet and compare with your prediction.

Q09 Variable Scope

Investigate access to variables inside and outside a block.

a. Declare `let city = "Delhi"` outside a block.

b. Inside a block, declare a different `let city = "Mumbai"`.

c. Print city inside and outside the block.

d. Declare a const only inside a block and try to access it outside.

e. Explain the resulting TypeScript error in your own words.

Q10 Debug the Program

Fix broken TypeScript statements.

a. Correct: `let 1name = "Amit";`

b. Correct: `const age = 20; age = 21;`

c. Correct the duplicate declaration: `let score = 10; let score = 20;`

d. Correct: `let first name = "Sam";`

e. Write one sentence explaining the reason for each correction.

Q11 Console Formatting

Practice clean output formatting with one console.log().

a. Print name and age separated by a space using commas.

b. Print name and age using string concatenation.

c. Print the same information using a template literal.

d. Use \n to display name, age and city on three different lines.

e. Explain the difference between a comma and a newline escape sequence.

Q12 Mini Project: Course Receipt

Build a small receipt generator using only variables and console output.

a. Create constants for course title, enrollment ID and course fee (Rs. 5999).

b. Create a let variable for discount starting at Rs. 500.

c. Calculate final payable fee.

d. Update discount to Rs. 750 and recalculate.

e. Print a multiline receipt with a heading, all amounts and final payable fee.

PART C / SHORT ANSWERS

Explain the why

• Why do programmers use variables instead of writing the same literal repeatedly?

• What is the difference between declaration, initialization and reassignment?

• When is const preferable to let? Why is var usually avoided in new code?

• What happens when a block-scoped variable is accessed outside its block?

• Why do we compile .ts files before running the generated .js files in this course?

FINAL SUBMISSION CHECKLIST

☐ Node.js, npm and TypeScript versions verified

☐ package.json and tsconfig.json created

☐ All 12 questions completed and commented

☐ Program compiled with npx tsc

☐ Output checked using node dist/assignment1.js

☐ Submitted source file and terminal screenshot